

Fast Minimization with MAGEMin & MAGEMinApp

Module 3 — Practical Thermodynamic Modelling for Petrology

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Objective

Use **MAGEMinApp**, a browser-based GUI for multi-core parallel Gibbs energy minimization, to build phase diagrams, melt/crystallization paths, and mineral-property maps — no coding required.

- Install & launch MAGEMinApp
- Setup: databases, bulk composition, diagram type
- Computing, reading, and exporting a phase diagram
- P–T–X paths and trace-element modelling

What is MAGEMin?

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Elevator pitch

A **parallel C library** that computes the thermodynamically most stable mineral assemblage for a given bulk-rock composition and P-T condition, using linear programming combined with an extended Partitioning Gibbs-Energy approach and gradient-based local minimization.

- Outputs: stable assemblage, phase proportions, melt fraction, density, seismic velocities
- Two interfaces: Julia (**MAGEMin_C**, scripted) or the browser (**MAGEMinApp**, point-and-click)

This workshop

We use **MAGEMinApp** exclusively — point-and-click in a web browser, no Julia scripting needed.

Why MAGEMin, and why now?

From Module 1

petropandas already exports bulk compositions in MAGEMin's input format.

Compared to Module 2

Where pypsbuilder builds one careful, topologically correct P–T pseudosection with THERMOCALC, MAGEMin **minimizes directly at every grid point** — much faster for large P–T–X grids, melting/crystallization paths, and geodynamic coupling, at the cost of not tracking Schreinemakers topology explicitly.

Installation

Installation: MAGEMinApp

Requires a recent Julia installation.

Launch Julia with multi-threading (use it — this is the whole point):

```
julia -t auto # or a specific count, e.g. julia -t 6
```

Install and launch the app:

```
using Pkg  
Pkg.add("MAGEMinApp")
```

```
using MAGEMinApp  
App()
```

Open `127.0.0.1:8050` in a web browser. MAGEMin_C installs automatically as a dependency — no separate step needed. Full details in the workshop's `install.md`.

Databases

Thermodynamic Databases

MAGEMin ships five built-in thermodynamic databases, selected by a short code when you initialize:

Code	System	Reference
ig	Igneous	Holland et al. (2018)
mp	Metapelite	White et al. (2014b)
mb	Metabasite	—
um	Ultramafic	—
Mantle	Mantle	—

Same underlying HPx-eos philosophy as the a-x models used by THERMOCALC / pypsbuilder in Module 2. MAGEMinApp exposes 13 database variants in total — including alkaline igneous and several mantle calibrations — selected from a dropdown in the Setup panel.

Caveat: each database is calibrated only over a limited P–T–composition range; calculations pushed outside it may fail or give unreliable results.

MAGEMinApp

Elevator pitch

An easy-to-use, web-based graphical interface to compute a whole range of phase diagrams efficiently — built on MAGEMin, no scripting required.

- P-T, T-X, P-X, and PT-X phase diagrams
- Polymetamorphic P-T paths, fractional melting / crystallization, isentropic paths
- Auto-labelled phase fields with iso-contours (phase fraction, density, seismic velocities)
- Export as vector graphics (SVG) or CSV data

Interface: Three Main Tabs

- **Phase Diagrams** — *Setup* (database, bulk composition, diagram type, P–T range) → *Diagram* (visualization & post-processing) → *Trace-elements* (partitioning & saturation)
- **PTX Path** — pressure–temperature–composition evolution: equilibrium, fractional melting, fractional crystallization, or isentropic mode
- **General Information** — reference tables for solution phases, end-members, bulk-rock input formats, and trace-element partition coefficients

Setting Up a Calculation

Setup panel

- Choose thermodynamic database and phase (solution) models
- Choose diagram type: P-T, T-X, P-X, PT-X, polymetamorphic
- Set P and T ranges, and a mesh-refinement level

Bulk-rock composition panel

- Upload a CSV/DAT file, or edit oxide values directly in the table
- Units: mol% or wt%; iron as FeO+O or FeO+Fe₂O₃
- Missing oxides are automatically treated as 0.0

Bulk-Rock Input Format

CSV header: metadata columns, then one column per oxide.

```
title,comments,db,sysUnit,SiO2,Al2O3,CaO,MgO,FeO,Fe2O3,K2O,Na2O,TiO2,H2O  
avgpelite,Module 1 export,mp,wt,58.98,16.06,1.75,4.15,7.30,,1.71,0.92,1.05,3.0
```

- Columns ending in `_frac2` give a second composition for P-X / T-X / PT-X diagrams
- Same idea as the `petropandas bulk-export` from Module 1 — just uploaded instead of pasted

Computing & Reading the Diagram

1. Click **Compute phase diagram** for the chosen P-T (or T-X/P-X) range
2. Phase fields are auto-labelled; colour by variance, a phase property, or oxygen fugacity
3. Apply adaptive mesh refinement near reaction boundaries for sharper lines
4. Click any grid point to read off mineral compositions and phase fractions
5. Add isopleths (phase mode, mineral composition, or a custom expression) with chosen contour intervals

Tutorial examples: KLB-1 (mantle), FPWorldMedian pelite, pyrolite, MORB, ijolite / Ne-syenite (igneous).

- **PTX Path tab:** equilibrium, fractional melting, fractional crystallization, or isentropic evolution along a P–T–X trajectory
- **Draw Path:** trace a P–T path by clicking points on a computed diagram; extract phase evolution along it
- **Trace elements:** partitioning & saturation modelling
- Oxygen-buffer calculations (QFM, NNO, ...) and latent heat of reaction
- Polymetamorphic paths and quantitative thermobarometry via Intersect

- Diagrams export as **SVG** — as one merged figure, or as individual layers (fields, lines, isopleths, labels)
- Numerical results export as **CSV** for further analysis (e.g. back into petropandas)

Bridge to Module 2

Compare the same bulk composition in MAGEMinApp and in your pypsbuilder pseudosection from this morning — do the stable assemblages agree?

Hands-on

Hands-on Exercise

1. Launch Julia with multiple threads and start the app: `julia -t auto` then `App()`
2. Upload (or paste) the bulk composition exported in Module 1
3. Compute a P–T diagram over a small range (e.g. 5–15 kbar, 500–900 °C) using the mp database
4. Add an isopleth and click a grid point to inspect phase fractions and compositions
5. Export the diagram as SVG

Recap

Recap & Resources

Key takeaways

- MAGEMin minimizes directly, point by point — fast enough for full grids and paths
- MAGEMinApp makes that speed accessible with no coding: Setup → compute → isopleths/paths → export, entirely in a web browser
- 13 thermodynamic database variants cover igneous, metapelitic, metabasitic, ultramafic, and mantle systems

Resources:

- github.com/ComputationalThermodynamics/MAGEMin
- github.com/ComputationalThermodynamics/MAGEMinApp.jl
- computationalthermodynamics.github.io/MAGEMin_C.jl/dev/MAGEMinApp
- Riel et al. (2022), *Geochemistry, Geophysics, Geosystems*, doi:10.1029/2022GC010427

End-to-End Workflow

Raw microprobe data (petropandas) → calibrated bulk composition → P-T pseudosection (pypsbuilder / MAGEMin) → P-T path

Questions & Discussion (16:30–17:00)